UK ONLINE RETAIL SALES ANALYSIS

Power Bi Portfolio Project

Data Analytics Project

Tools Used:

• Microsoft Excel • Power Query • Power BI

REPORT

1. PROJECT OVERVIEW

This project analyzes UK online retail sales data to understand the company's sales performance, revenue trends, customer markets, and product performance.

Using Power Query, I cleaned and prepared the raw transaction data before analyzing it in Power BI. I created key calculations and an interactive dashboard to explore overall sales performance, monthly revenue trends, country-level performance, and the products generating the most revenue.

The analysis provides data-driven insights that can help the business understand its strongest markets, identify sales patterns, and make informed decisions to support revenue growth.

2. PROJECT OBJECTIVES

The main objectives of this project were to:

- **Assess overall sales performance** using key sales and order metrics.
- **Analyze monthly revenue trends** to understand how sales changed over the period.
- **Evaluate country-level performance** to identify the company's strongest markets.
- **Identify top-performing products** based on revenue generated.
- **Provide data-driven insights and recommendations** that can support business decision-making and future sales growth.

3. BUSINESS PROBLEM

- The company has a large volume of online retail transaction data, but without proper analysis, it can be difficult to clearly understand its sales performance, revenue patterns, market performance, and product contribution.
- The business therefore needed to transform its raw transaction data into meaningful information that could help management identify sales trends, understand the strongest markets, evaluate product performance, and make informed decisions to support future growth.

4. KEY BUSINESS QUESTIONS

The analysis was guided by the following key business questions:

1. **How is the business performing overall?**
2. **How does revenue change from month to month?**
3. **Which countries generate the most revenue?**

4. **Which countries have the highest quantity of products sold?**
5. **Which products generate the most revenue?**

5. DATA SOURCE AND DESCRIPTION

Source

The dataset was obtained from the **UCI Machine Learning Repository**. It contains transaction data from a UK-based online retail business covering the period from **December 2010 to December 2011**.

The raw dataset contained **541,909 rows and 8 columns**.

Description

The dataset contains information about customer transactions, products, sales quantities, prices, transaction dates, and customer locations.

- **InvoiceNo:** Unique invoice or transaction number.
- **StockCode:** Product identification code.
- **Description:** Description of the product purchased.
- **Quantity:** Number of units purchased.
- **InvoiceDate:** Date and time when the transaction occurred.
- **UnitPrice:** Price charged for each unit of the product.
- **CustomerID:** Identification number of the customer.
- **Country:** Country where the customer was located.

6. DATA CLEANING & PREPARATION

The raw dataset was cleaned and prepared using **Power Query** before conducting the analysis. During the cleaning process, several data-quality issues were identified and addressed:

- **Cancellations:** Cancellation transactions were identified and removed.
- **Negative quantities:** Transactions with negative quantities were removed.
- **Duplicate rows:** Duplicate transaction records were removed.
- **Null descriptions:** Records with missing product descriptions were removed.
- **Null unit prices:** Records with missing unit prices were removed.
- **Negative unit prices:** Records with negative unit prices were removed.

- **Zero-unit prices:** These records were investigated and retained after determining that they should not be removed from the analysis.
- **Missing Customer IDs:** Missing CustomerID values were retained and replaced with **"Unknown"** so that the records could still be included in the analysis.

The cleaned dataset was then used for the sales analysis and Power BI dashboard.

7. ANALYSIS & CALCULATIONS

After cleaning the dataset in Power Query, I loaded the cleaned data into **Power BI** for analysis. I created calculated columns and measures to support the analysis, build the dashboard, and answer the key business questions.

Calculated columns created:

- **Sales Value** – calculated from the quantity purchased and unit price for each transaction.
- **Year** – extracted from the invoice date.
- **Month** – extracted from the invoice date.
- **Month Number** – used to arrange the months in chronological order.

Measures created to evaluate overall business performance:

- **Total Revenue** – the overall revenue generated from sales within the selected period.
- **Average Sales Value** – the average value of individual sales lines.
- **Count of Orders** – the number of distinct orders/transactions.
- **Average Amount per Order** – the average revenue generated per order.
- **Total Quantity** – the total number of products sold.

These calculations provided the key performance indicators needed to understand overall sales performance and answer the business questions through the Power BI dashboard.

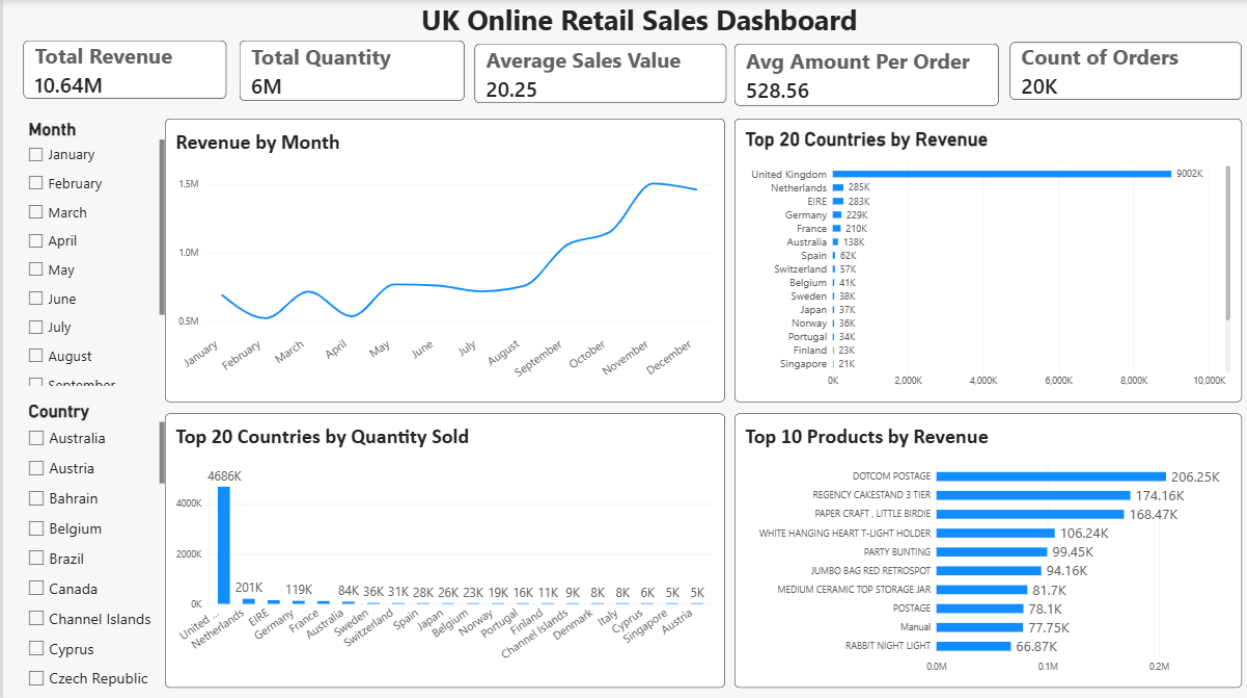
8. POWER BI DASHBOARD

An interactive **Power BI dashboard** was developed to present the results of the analysis in a clear and user-friendly format.

The dashboard includes key performance indicators for **Total Revenue, Total Quantity, Average Sales Value, Average Amount per Order, and Count of Orders.** Interactive Country and Month slicers allow users to explore the results based on their selected filters.

The dashboard also presents revenue and sales patterns through visualizations showing top countries by revenue, top countries by quantity sold, top products by revenue, and monthly revenue trends.

This interactive dashboard enables management to quickly explore sales performance and identify key patterns across markets, products, and time periods.



9. KEY FINDINGS

1. Overall Business Performance

The analysis provides an overview of the UK online retail business's sales performance during the period covered by the dataset.

The business generated approximately **£10.64 million in total revenue** from approximately **20,000 orders**, with around **6 million units sold**. The **average sales value was £20.25 per sales line**, while the **average amount per order was £528.56**.

These figures provide a baseline view of the company's sales activity during the period. However, because the dataset does not provide business targets, budgets, or previous-period benchmarks, the analysis cannot determine whether the performance was above or below the company's target. The figures therefore describe the observed performance rather than judging it as good or poor.

2. Monthly Revenue Trend

The analysis shows that monthly revenue generally increased over the period, although there were some fluctuations between months.

November recorded the highest monthly revenue at approximately £1.5 million, while **February recorded the lowest at approximately £522,000**.

This indicates that sales performance varied considerably across the months, with revenue reaching its highest level toward the end of the period.

3. Revenue by Country

The **United Kingdom generated the highest revenue**, with total revenue of approximately **£9.0 million**.

There was a significant difference in revenue between the United Kingdom and the Netherlands, which ranked second. This shows that the United Kingdom was by far the largest revenue-generating country in the dataset.

4. Quantity Sold by Country

The **United Kingdom recorded the highest quantity sold, at approximately 4.6 million units**. The other countries had significantly lower quantities, with the quantities for the other countries **below 250,000 units**.

5. Top Products by Revenue

Dotcom Postage led the ranking, generating approximately **£206,000 in revenue**, followed by **Regency Cake Stand 3 Tier** at approximately **£174,000**. The remaining top-performing products recorded relatively similar revenue levels.

10. INSIGHTS

1. Overall Business Performance

The figures show that the business handled a substantial volume of sales activity, generating approximately £10.64 million in revenue from around 20,000 orders and 6 million units sold during the period analyzed.

2. Monthly Revenue Trend

Revenue varied across the months, with stronger sales occurring toward the end of the period.

3. Revenue by Country

The business is highly concentrated in the UK market, meaning the UK is the company's most important revenue-generating market in this dataset.

4. Quantity Sold by Country

The UK accounts for the vast majority of units sold, indicating that sales volume is heavily concentrated in this market.

5. Top Products by Revenue

Revenue varied across the top-performing products, with no single product accounting for a dominant share of revenue.

11. RECOMMENDATIONS

1. UK Market

The company should continue to support and strengthen its UK market while exploring opportunities to increase sales and expand its customer base in other countries.

2. Monthly Revenue

The company should develop strategies to strengthen sales throughout the year, particularly during lower-revenue months, rather than relying mainly on stronger sales toward the end of the year.

3. Product Performance

The company should continue to monitor and promote strong-performing products while identifying opportunities to improve the performance of other products.

12. CONCLUSION

This project transformed raw UK online retail transaction data into a structured and interactive Power BI dashboard. Through data cleaning, analysis, and visualization, the project provided a clearer understanding of overall sales performance, monthly revenue patterns, country-level performance, and product revenue.

The analysis highlighted the strong contribution of the UK market, variations in monthly revenue, and differences in product performance. These findings provide useful information that can support data-driven decision-making and help the business identify opportunities for strengthening sales and expanding its customer base.